Flavours of the future: The case of Wolof *di*

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July 11, 2025

Abstract: The Wolof auxiliary *di*, which has been glossed as ‘imperfective’ in previous descriptive work, can be used to express event-in-progress, habitual and future meanings. However, the availability of these readings depends on the structural position of *di* within the clause. In a low clausal position (below T), all three readings for *di* are available, however in a high clausal position (above T), only a future reading is available. Furthermore, we observe distinct “flavours” of future interpretation for *di* depending on its syntactic position. For low *di*, a prediction is made based on preparatory stages that are already in motion, whereas for high *di*, a more general prediction is made. We analyze *di* as an event-relative modal, following previous literature designed to capture (i) restrictions on modal flavour based on syntactic position (Hacquard, 2006, 2010), and (ii) modal properties of progressives and habituals (Portner, 1998; Ferreira, 2016). Our analysis derives the restrictions on the interpretations of *di*, and bears on debates surrounding the modal flavours of future expressions cross-linguistically.

1 The puzzle

The auxiliary *di* in Wolof (West Atlantic Niger-Congo), glossed as ‘imperfective’ in descriptive sources (*inaccompli* in the literature in French) (e.g., Church 1981; Robert 1991; Diouf 2003; Voisin 2010) is associated with several readings. First, an event-in-progress reading is available as in (1), as well as a habitual reading as in (2). It is based on the availability of these readings that *di* is often considered to be an imperfective marker. In addition to these imperfective meanings, *di* can also convey a future meaning, as in (3).[1]

(1) **Progressive reading; low *di***
Daf-a=∅ **di** (> dafay) añ, mën-ul ñëw.
do-C=3SG.S DI eat.lunch can-NEG come
‘Il est en train de manger, il ne peut pas venir.’/
‘He is eating, he cannot come.’ (Robert, 1991, p. 263)

(2) **Habitual reading; low *di***
Daf-a=∅ **di** (> dafay) jaay.
do-C=3SG.S DI sell

[1] Data from Robert 1991 use translations and context descriptions in French. The glosses and English translations are our own. Throughout, we gloss *di* as DI so as not to presuppose an analysis before arguing for our own. Glosses otherwise follow the Leipzig Glossing Rules with the following exceptions: C = low complementizer, HABIT = habitual, O = object, S = subject. Note that *di* is pronounced as a glide when it follows clitics; the pronunciation of the C-cluster together with clitics and *di* is given in parentheses, e.g. (>dafay) in (1). Examples not otherwise marked are from Martinović’s fieldwork.

‘Il vend.’ = ‘Il est marchand.’/‘He sells’ = ‘He’s a merchant.’ (Robert, 1991, p. 267)

(3) **Future reading; low *di***
Context: devant la maison en construction/in front of a house under construction
Kii mu-a **di** (> mooy) rafet kër!
this.one 3SG.S-C DI be.pretty house
‘Elle va être drôlement belle, sa maison, à lui!’/‘It’s going to be really beautiful, this one’s house.’ (Robert, 1991, p. 269)

Interestingly, the availability of these readings depends on *di*’s structural position. In (1)-(3), when *di* is in a “low” position (below C, as will be argued below), all three readings—in progress, habitual, and future—are available. When *di* is in a higher position, in C, only the future is possible, as in (4). As we will show later, when *di* is part of the C-complex, the event-in-progress and habitual readings are not available, which is already hinted at by the comment from Robert that (4) cannot be used if the event is in progress. Throughout this article, we use the descriptive term “low *di*” when *di* is structurally lower than C, and “high *di*” when *di* is in C.

(4) **Future reading; high *di***
Di-na=∅ gor garab bi.
DI-C=3SG.S cut tree the
‘(À ce moment là) il abattra l’arbre.’/‘(At that time) He’s going to cut the tree.’
[*“impossible s’il est en train d’essayer de l’abattre”/impossible if he is already trying to cut it*] (Robert, 1991, p. 272)

A closer look at the future readings reveals that the sentences with low vs. high *di* convey different “flavors” of future reference. For instance, (5) is a prediction that Fatou’s son will be king based on the fact that Fatou is the queen, while (6) is a prediction based on a dream. Speakers are quite clear that these sentences cannot be interchanged in these contexts, and the judgments are robust.[2]

(5) CONTEXT: Fatou is a queen, so her son will automatically become the king one day.
Faatu xam-na-∅ ni doom-om ba mu juddoo daf-a=∅ di (>dafay)
Fatou know-C=3SG.S that child-POSS.3SG when 3SG.S be.born do-C=3SG.S DI
nekk buur.
be king
‘Fatou knows that her son, when he is born, is going to be king.’ (low *di*)

(6) CONTEXT: Fatou is poor, but she had a dream that she will have a son who will become king.
Faatu xam-na=∅ ni doom-om ba mu juddoo di-na=∅ nekk buur
Fatou know-C=3SG.S that child-POSS.3SG when 3SG.S be.born DI-C=3SG.S be king
‘Fatou knows that her son, when he is born, is going to be king.’ (high *di*)

Robert (1991) took the different readings of *di* in the different positions as evidence for two distinct lexical items. She furthermore does not distinguish between the different future readings available for high vs. low *di* in (5)-(6). We wish to provide a unified analysis of the readings of *di* by combining and expanding on several independently motivated analyses for progressives, habituals, and modality in the literature. In a nutshell, we treat *di* as an event-relative modal operator, following the analysis of Portner (1998) for the English progressive, and Hacquard (2010)

[2] We use *will* or *be going to* in our English translations of Wolof sentences with future temporal reference, without a commitment to the semantics of English *will* or *be going to*.

for modal auxiliaries in English and other languages. Hacquard's analysis is specifically designed to derive distinct modal flavours from modals that appear in different structural positions, with epistemic readings associated with a high position and circumstantial readings associated with a low position. Evidence for the correlation between structural position and modal flavour is indirect in English, but we argue that the behaviour of Wolof *di* provides new overt evidence for this view (Hacquard, 2010; Kush, 2011). Progressives and habituals cross-linguistically have also been argued to involve event-relative circumstantial modality (Portner, 1998; Ferreira, 2016). We adopt this view to explain why in-progress and habitual readings are only available for *di* in its low position. Futures in Wolof are derived from either a circumstantial or epistemic modal base, depending on its structural position, and give rise to different readings, as in (5)-(6). Our analysis is decompositional, in that we assign only a modal semantics to *di*, and factor out the aspectual components to null imperfective and prospective operators that must be licensed by *di*.

This paper proceeds as follows. In sections 2 and 3, we present some background on Wolof and the syntax and interpretation of the auxiliary *di*. Then, in section 4 we introduce the key aspect of our analysis – event-relative modality – by reviewing Portner's (1998) analysis of the progressive and Ferreira's (2016) analysis of habituals, applying them to the progressive and habitual readings of *di*. We then turn to the future readings of *di* in section 5, and motivate the idea that the future interpretations of high vs. low *di* are a derived from epistemic vs. circumstantial modal bases, respectively. In section 6 we lay out our full analysis of high and low *di*, which derives the differences in modal flavour from *di*'s structural position. In section 7 we compare the behaviour of *di* to other modal verbs in Wolof, which do not display interpretational differences based on structural position, and we propose that other modals in the language are not event-relative like *di*. We also briefly explore the occurrence of *di* in the complements of control predicates. Section 8 concludes.

2 Background on Wolof and syntax of *di*

Wolof is a West-Atlantic Niger-Congo language spoken in Senegal, the Gambia, and parts of Mauritania. This paper is concerned with the dialect of Wolof spoken in northern Senegal, specifically in St.-Louis (Ndar) and the Walo region. Unless otherwise noted, all data in this paper come from original fieldwork in the city of St.-Louis and the village of Ndombo Alarba between 2016 and 2019, using French as the mediating language. Data collection involved direct elicitation, using a mix of translation tasks, judgment tasks, and responses to targeted storyboards (Matthewson, 2004; Burton & Matthewson, 2015; Bochnak & Matthewson, 2020). The reported judgments are robust and stable across speakers.

Wolof is an optional tense language, with one optional past tense morpheme, *oon* (Bochnak & Martinović 2017). Eventive predicates with no tense or aspect morphemes receive an episodic, default past interpretation, as in (7). Stative predicates have a default present reading, shown in (8).

(7) **Episodic reading with eventive predicates**
Xale yi lekk-na=ñu ceeb.
child the.PL eat-C=3PL.S rice
'The children ate rice.'

(8) **Present reading with stative predicates**
Mbaye bég-na=∅.
Mbaye be.happy-C=3SG.S
'Mbaye is happy.'

Finite indicative clauses in Wolof all have one of a series of *sentence particles*, such as *na* in (7) and (8). Following Dunigan (1994) and Martinović (2015, 2023), we consider these to all be contained in one high head that Dunigan labels Sigma in the spirit of Laka (1990), and Martinović a left-peripheral low C-head; we follow the latter analysis. The CP-layer is obligatory for the sentence to have independent temporal interpretation, and inflectional elements such as negation and the optional tense morpheme only occur in the presence of C (Njie 1982; Zribi-Hertz & Diagne 2003). Wolof also has clauses with no sentence particles which can occur unembedded, as in (9). Such clauses cannot contain any inflectional material—tense, aspect, or negation—and must obtain temporal reference from context or a directly preceding temporal adverb (Sauvageot 1965; Church 1981; Dialo 1981; Robert 1991; Zribi-Hertz & Diagne 2002). Finally, sentence particles, negation and tense are also obligatorily absent from embedded non-finite clauses, as in (10). Therefore, to the extent that the notion of *finiteness* is represented in Wolof, it is tied to the presence of a sentence particle.

(9) Demba ak Fama tabax kër.
Demba and Fama build house
'Demba and Fama are building/build/will build a house.'

(10) Demba ak Fama bëgg-na=ñu [Musaa tabax kër].
Demba and Fama want-C=3PL.S Moussa build house
'Demba and Fama want for Moussa to build a house.'

In this paper, we are mainly interested in finite declarative clauses with the sentence particle *na*, as in (11), and the sentence particle *da(f)*, in (12). In (11), the highest verbal element in the clause is found in C, here the main verb *tabax* 'build'. In (12), the verb remains below C. Senghor (1963) proposes that the sentence particle in such clauses contains the verb *def* 'do', based on Wolof dialects where the presence of *def* is more transparent; we therefore gloss this C as "do.C" to make the distinction between the two clauses clear, but nothing hinges on this particular analysis of the sentence particle. In both of these structures the clause-internal subject is obligatorily a clitic (*ñu* in the examples below), which surfaces right-adjacent to C, where all clitics in finite clauses cluster in Wolof (Russell 2006; Martinović to appear). An optional non-clitic subject is found to the left of C.

(11) **Neutral declarative**
Demba ak Faatu **tabax**-$\boxed{\text{na}}$=ñu kër.
Demba and Fatou build-C=3PL.S house
'Demba and Fatou built a house.'

(12) **Predicate Focus**
Demba ak Faatu $\boxed{\text{da}}$=ñu **tabax** kër.
Demba and Fatou do.C=3PL.S build house
'Demba and Fatou BUILT A HOUSE.'

The meaning difference between clauses as in (11) and those in (12) is roughly between information-structurally neutral sentences, which are felicitous in an out-of-the-blue context, and predicate focus. With eventive verbs, structures with *da(f)* are not felicitous in an out-of-the-blue context and generally involve either V or VP focus, as indicated in the translation (where capitalization indicates focus), or are used in contexts that the descriptive or functionalist literature calls 'explicative' (e.g. Robert 1991); for instance, the sentence in (12) may be used as an explanation for why Demba and Fatou are throwing a party. With stative verbs, however, this contrast between (11) and (12) disappears, and both structures can be used in an out-of-the-blue context, and many speakers show preference for the *da(f)*-clauses.

Our focus in this paper are structures that contain the auxiliary element *di*. In *na*-sentences, *di* occupies the position of the highest verbal element in the clause, which is found in C, shown in (13). In *da(f)*-sentences, *di* remains below C, as in (14).

(13) ***Di* in a *na*-sentence**
Demba ak Fama **di**-na=ñu tabax kër.
Demba and Fama DI-C=3PL.S build house
'Demba and Fama will build a house.'

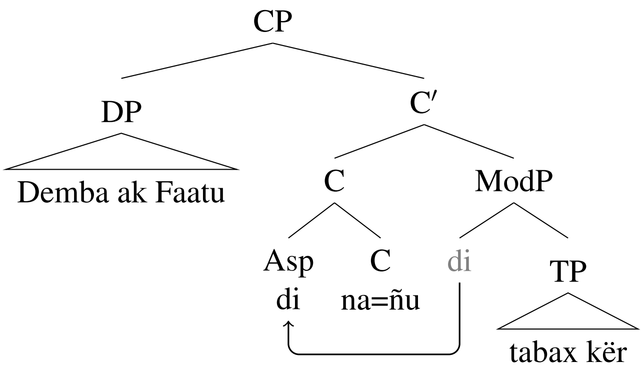
(14) ***Di* in a *da(f)*-sentence**
Demba ak Fatou $\boxed{\text{da}}$=ñu **di** (> dañu**y**) tabax ay kër.
Demba and Fatou do.C=3SG.S DI build INDF.PL house
'Demba and Fatou will build houses/are building houses/build houses.'

When nothing is suffixed onto it, *di* cliticizes onto the preceding phonological word and is pronounced as the glide [j] (***-y*** in the orthography), which is one of the reasons some authors (e.g., Robert 1991) consider the *di* that surfaces in C and the one that is below C two distinct elements. Evidence that surface phonological form should not be used as a metric for distinguishing morphosyntactic categories comes from examples where *di* is suffixed with another element, for example negation, in (15). In that case, the consonant *d* again surfaces.

(15) Demba ak Fatou da=ñu **d(i)-ul** tabax ay kër.
Demba and Fatou do.C=3PL.S DI-NEG build INDF.PL house
'Demba and Fatou won't build houses/aren't building houses/don't build houses.'

Our analysis treats *di* that surfaces in C and the one below C as one and the same modal element, which can be merged in two different syntactic positions. In this respect, we follow Hacquard (2006, 2010) in allowing modals to occupy positions above and below T. Specifically, we argue that *di* is a modal head, which in a *na*-sentence, as in (13), is merged above the TP, and moves to C, illustrated in (16). We henceforth refer to this as 'high *di*'.[3]

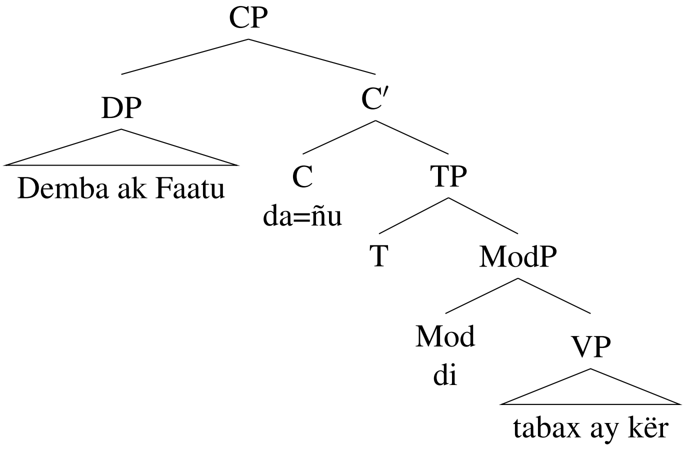
(16)



In sentences where *di* does not move to C, as in (14), we propose that it is generated below T, shown in (17). We refer to this as 'low *di*'. Note that low *di* may move to T; this is not relevant for our analysis.

[3]For a justification of other elements of the clausal structure of Wolof that we follow here, e.g., the position of the non-clitic subject in Spec,CP, see Martinović 2015, 2023.

(17)



For now, we stipulate that 'low' *di* may not be merged in *na*-sentences, and that 'high' *di* may not be merged in *da(f)*-sentences.

In the following section we show that different readings of *di* are available in the two positions. First, below C, *di* can have event-in-progress, habitual, or future readings. In C, only the future reading is available. Second, the future reading has different 'flavors' in the two positions; in the lower position, future *di* describes a prediction based on circumstances related to the VP event (in a way to be made precise later), whereas in the high position, future *di* makes a prediction without taking such circumstances into account. We argue that both generalizations can be accounted for by analyzing *di* as an event-relative modal (Hacquard, 2006, 2010), where the modal "flavor" of *di* is determined by the nature of the event variable it takes as an argument. The aspectual properties associated with the various readings of *di* are factored out into aspectual heads that co-occur with it.

3 The readings of *di* in low and high positions

In this section, we illustrate the differences in meanings that *di* can have depending on its syntactic position. We show that progressive and habitual readings are only possible with low *di* while a future reading is possible for both high and low *di*. Later, in section 5, we discuss the differences in the future interpretations of high versus low *di*.

With low *di*, we find that progressive and habitual readings are available. Examples of the progressive reading are shown in (18) and (19). In (18), the event of drawing a circle has already started and is ongoing at the speech time. (19) contains an achievement predicate *tisooli* 'sneeze', and an iterative interpretation obtains. Assuming that achievements are viewed as instantaneous events (Bennett & Partee, 1978), such an event cannot be ongoing at a reference time. To circumvent this restriction, the event is then construed as the sum of several sneezing sub-events occurring in succession.

(18) CONTEXT: I am standing in front of a wall with a bucket of paint and I just put a brush to the wall and started drawing something. Someone walks into the room and asks *What are you doing?* I respond:
Da=ma di (> damay) rëdd wërëngërël.
do.C=1SG.S DI draw circle
'I am drawing a circle.'

(19) CONTEXT: I hear a repetitive noise from another room, and I ask what that is. Someone answers me:
Dudu daf-a=∅ di (> dafay) tisooli.
Dudu do-C=3SG.S DI sneeze
'Dudu is sneezing.'

The habitual reading of low *di* is illustrated in (20) with an accomplishment predicate, and in (21) with a stative predicate.

(20) CONTEXT: My friend Fatou goes around and draws circles on walls every Monday. Another friend has seen her a few times walking around with a bucket of paint, and asks me what that's about. I tell him:
Altine bu nekk, Faatu daf-a=∅ di (>dafay) rëdd ay wërëngërël.
Monday C$_{\text{REL}}$ be, Fatou do-C=3SG.S DI draw INDF.PL circle
'Every Monday, Fatou draws circles.'

(21) CONTEXT: Magatte visits her village rarely, only once every two years, but whenever she goes there, her friend Binta is pregnant. Magatte comments to her mother:
Binta daf-a=∅ di (>dafay) ëmb rekk!
Binta do-C=3SG.S DI be.pregnant only
'Binta is always pregnant!'

Low *di* can also express future readings, as in (22).

(22) CONTEXT: I see Mbaye walking around the town with an architect, buying building materials, etc., and I ask our mutual friend what Mbaye is up to, and he tells me:
Mbaye daf-a=∅ di (> dafay) tabax kër.
Mbaye do-C=3SG.S DI build house
'Mbaye is going to build a house.'

A future reading also obtains with high *di*, as in (23).

(23) Mbaye di-na=∅ tabax kër.
Mbaye DI-C=3SG.S build house
'Mbaye is going to build a house.'

Both low *di* and high *di* can also express future meanings with stative predicates, as in the examples in (24) and (25). (26) illustrates a future meaning with an achievement predicate.

(24) CONTEXT: Fanta and her husband Ibrahim cannot conceive a child, so they go to see a 'doctor' who uses local plants and herbs to make medicine. He gives them a tea and tells Fanta to drink it every day, promising:
Fanta daf-a=∅ di (>dafay) ëmb.
Fanta do-C=3SG.S DI be.pregnant
'Fanta is going to get pregnant.'

(25) CONTEXT: Mbaye is making his wife Magatte's favorite dish for lunch and I comment to him:
Magatt di-na=∅ bég.
Magatte DI-C=3SG.S happy
'Magatte is going to be happy.'

(26) CONTEXT: I am throwing a party and I ask Fatou if anyone from her family will be there. She tells me that her brother Moussa will come:

Musaa di-na=∅ ñëw.
Moussa DI-C=3SG.S come
'Moussa will/is going to come.'

The key difference between the low *di* and the high *di* is that the former can express progressive and habitual readings, in addition to future readings, while the latter can only express future readings. This contrast is illustrated in (27).

(27) CONTEXT: There is a party and Magatte is dancing. Her husband Mbaye does not like it when she dances in public, so he is in a bad mood. A friend asks what is wrong, and Mbaye says:

a. Magatte daf-a=∅ di (> dafay) fecc.
Magatte do-C=3SG.S DI dance
'Magatte is dancing.'/'Magatte always dances.'

b. #Magatte di-na=∅ fecc.
Magatte DI-C=3SG.S dance
Intended: *'Magatte is dancing.'/'Magatte always dances.'*

The availability of the three readings of *di* in the high and the low position are summarized in (28).

(28) The available readings of high *di* and low *di*

	progressive	habitual	future
low *di*	✓	✓	✓
high *di*	#	#	✓

4 Towards an analysis of progressive and habitual readings

In this section, we sketch the basic ingredients of the progressive and habitual readings of *di*. In Section 5, we discuss the future readings in more detail, before presenting our unified analysis in Section 6.

Following much previous work, we take it that imperfective meanings (e.g., progressive, habitual) incorporate a modal component (Dowty, 1979; Landmann, 1992; Portner, 1998; Arregui et al., 2014; Ferreira, 2016; Cable, 2022). Furthermore, we follow Portner (1998) that the modality involved is *event-relative*: in contrast to Kratzer's (1981, 2012) analysis of modal auxiliaries where the modal possibilities are projected from a world variable, the modal possibilities of imperfective meanings are projected from an event variable (see also Hacquard 2006, 2010 for modal auxiliaires). Specifically, the event variable associated with the event description in the verb phrase projects a set of possibilities that are compatible with the circumstances surrounding that event in a way to be explained below.

Let us first introduce the main components of the semantics of modal auxiliaries following Kratzer (1981, 2012), which invokes quantification over possible worlds. The two main components of the meaning of modal auxiliaries such as *must* and *can* are known as FLAVOUR and FORCE. The two main forces that are distinguished by modal auxiliaries in many languages are possibility and necessity. The former involves existential quantification over a set of worlds, and the latter universal quantification. In English, *can* is an example of a possibility modal (in some relevant worlds, the embedded proposition is true), while *must* is a necessity modal (in all relevant worlds, the embedded proposition is true).

Modal flavour relates to what counts as a relevant set of worlds that the modal quantifies over. For example, an epistemic modal claim relates to a body of knowledge from which a conclusion is drawn, while a deontic modal claim relates to a set of rules that hold in the evaluation world. In English and many other languages, a single modal may be compatible with different flavours, as shown in (29) below for *must*.

(29) a. Fatou must be in her office, because the light is on. (epistemic)
b. Fatou must be in her office between 9am and 5pm on weekdays. (deontic)

In Kratzer's analysis of modal auxiliaries, the quantificational domain of a modal is derived from two parameters, known as a MODAL BASE and an ORDERING SOURCE. These both serve as semantic arguments of the modal, and their identity is derived from context, or for some modals may be lexically specified. The modal base is derived from an accessibility relation f, which relates to a world w a set of propositions that are true in worlds accessible from w based on f. For an epistemic modal, this would be a set of propositions that the speaker knows to be true in w that serve as evidence for the embedded proposition. For a circumstantial modal, this would be a set of circumstances in w. Then, $\cap f(w)$ is the set of worlds compatible with what the speaker knows in w or worlds compatible with the circumstances in w, for an epistemic or circumstantial modal, respectively. This set of worlds can then be ordered by the ordering source g, another function on worlds w that returns a set of propositions. The set of propositions corresponds to a set of laws or rules in w for a deontic modal, or a set of goals in w for a teleological modal, for instance. The propositions in $g(w)$ induce an ordering $<_{g(w)}$ on the worlds in $\cap f(w)$ according to how many propositions in $g(w)$ are true in each of the worlds in $\cap f(w)$. The relation $w' <_{g(w)} w''$ holds if and only if more propositions in $g(w)$ hold in w' than in w''. In this sense, the world w' is said to be better than the world w'' according to the ordering source. So for a deontic ordering source, w' would be a better world than w'' if and only if more of the laws of w are followed in w' than in w''. A modal then quantifies over a subset of the worlds in $\cap f(w)$, which, following Portner (1998) we refer to as the "best" worlds, which we will represent using the notation $\text{BEST}(f, g, w)$. This set consists of worlds in $\cap f(w)$ that are ideal according to $g(w)$.

Putting this all together, a necessity modal says that the embedded proposition is true in all of the best worlds in the modal base according to the ordering source, while a possibility modal says that embedded proposition is true in at least one of the best worlds in the modal base according to the ordering source. This is sketched in (30), where p is the embedded proposition, and the modal flavour is derived based on the identities of f and g given by the context.

(30) a. $[\![\textit{must}]\!] = \lambda f_{\langle s,\langle st,t\rangle\rangle} \lambda g_{\langle s,\langle st,t\rangle\rangle} \lambda p_{\langle s,t\rangle} \lambda w_s . \forall w' \in \text{BEST}(f, g, w) : p(w') = 1$
b. $[\![\textit{can}]\!] = \lambda f_{\langle s,\langle st,t\rangle\rangle} \lambda g_{\langle s,\langle st,t\rangle\rangle} \lambda p_{\langle s,t\rangle} \lambda w_s . \exists w' \in \text{BEST}(f, g, w) : p(w') = 1$

To get our analysis for the Wolof auxiliary *di* off the ground, we will begin by looking at the progressive reading. We will closely follow Portner's (1998) analysis of the English progressive. Portner's modal account of the progressive uses the main insights of Kratzer's analysis of modal auxiliaries, namely that its interpretation is relative to a modal base and ordering source. The twist introduced by Portner for the progressive is that the modal base and ordering source are both projected from an event as opposed to a world of evaluation, i.e., the functions f and g take an event e as an argument instead of a world w.

Let us illustrate how we can apply Portner's analysis using example (18), repeated here in (31).

(31) Da=ma di (> damay) rëdd wërëngërël.
do.C=1SG.S DI draw circle
'I am drawing a circle.'

The modal base is circumstantial, comprising the set of circumstances relevant to whether an event e is completed. For (31), $f(e)$ would consist of propositions such as 'I intend to draw a circle', 'I know how to draw a circle', 'I am paying attention to the task', etc. The ordering source is what Portner calls "non-interruption," which contains propositions that assert or entail that the event does not get interrupted. For (31), $g(e)$ would consist of propositions such as 'My paintbrush doesn't break', 'I don't run out of paint', 'I don't get distracted', etc.

Portner notes that the modal base and ordering source must be sensitive not only to the event e but also to the event description given by the verb phrase. This is because one and the same event can be described in different ways, and the progressive is sensitive to these differences. It is from the linguistically given event description that the participants of the event become relevant, and may impact the identity of the modal base and ordering source. We will thus add an event description P as an argument of the modal base and ordering source, and notate the circumstantial modal base and non-interruption ordering source, respectively, relative to an event e and event description P as $\textsc{circ}(e, P)$ and $\textsc{ni}(e, P)$, as in (32).

(32) a. modal base: $\textsc{circ}(e, P)$ = {'I intend to draw a circle', 'I know how to draw a circle', 'I am paying attention to the task', ...}
b. ordering source: $\textsc{ni}(e, P)$: {'My paintbrush doesn't break', 'I don't run out of paint', 'I don't get distracted', ...}

Like other modal operators, the progressive quantifies over a subset of worlds given by the modal base, which are in some sense ideal with respect to the ordering source. Portner calls these the inertia worlds: those in the circumstantial modal base where the event e is not interrupted. Formally, the worlds in $\cap\textsc{circ}(e, P)$ are ordered with respect to how many propositions in $\textsc{ni}(e, P)$ are true in them. The progressive quantifies only over the best worlds: those in the circumstantial modal base where *all* the propositions in the non-interruption ordering source are true. This set is represented by BEST(CIRC,NI,e,P), as in (33), where $<_{\mathrm{NI}}$ is an ordering between worlds relative to the non-interruption ordering source. That is, $w' <_{\mathrm{NI}} w$ means that w' is better than w with respect to the non-interruption ordering source. The progressive operator then quantifies over that set.

(33) BEST(CIRC,NI,e,P) = $\{w \in \cap\textsc{circ}(e, P) | \neg\exists w' \in \cap\textsc{circ}(e, P)$ s.t. $w' <_{\mathrm{NI}} w\}$

This set-up accounts for the possible non-actualization of a P-type event in the evaluation world, or what is known as the imperfective paradox for telic predicates. If in the evaluation world the event gets interrupted, then it is not a member of the set BEST(CIRC,NI,e,P), and is therefore not part of the domain of quantification of the progressive. The general template for the modal component of the progressive – and what we propose for Wolof *di* – is shown in (34). The step of adding a temporal component to this will be taken up immediately below.[4]

(34) Modal component of *di* applied to an event predicate P and evaluation world w:
$\exists e[\forall w' \in$BEST(CIRC, NI,$e)[\exists e'[P(e')(w')]]]$

The progressive additionally has an important temporal component associated with it, namely that an event e is ongoing at the reference time t. This is an essential part of progressive event descriptions, in contrast to perfective ones, the latter of which place the totality of an event within the reference time. Crucially, e as an event associated with event description P does not culminate within the reference time, and indeed might not even culminate in the evaluation world, as explained above. Rather, in all the best worlds, there is a time t' of which the reference time t is a non-final sub-interval, where t' is the run-time of a P-event e'. That is, the event e' culminates in

[4]To avoid cluttering up the truth conditions, going forward we will suppress P from the specification of BEST.

the future of the reference time t in all the inertia worlds.

Putting the modal and temporal pieces together, a Portner-style analysis of the truth conditions of the progressive reading of (31), repeated below as (35a), is shown in (35b). In prose, (35a) is true if and only if there exists an event e that is ongoing at the reference time t, and that in all the inertia worlds w', there exists an event e' and time t' such that t' is the run-time of e', t is a non-final interval of t', and e' is a circle-drawing event by the speaker that takes place in w'.

(35) a. Da=ma di (> damay) rëdd wërëngërël.
do.C=1SG.S DI draw circle
'I am drawing a circle.'

b. $[\![$*damay rëdd wërëngërël*$]\!]$
$= 1$ iff $\exists e \exists t[t \subseteq \tau(e)$ & $\forall w' \in \text{BEST}(\text{CIRC}, \text{NI}, e)[\exists e' \exists t'[t \subset_{nf} t'$ & $t' = \tau(e')$
& $\mathbf{draw}(sp, c, e', w')]]]$

Turning to the habitual reading of low-*di*, we follow Ferreira (2016), who argues that the basic analysis set up by Portner for progressives can be modified slightly to account for habituals as well. Ferreira argues that it is desirable to have a unified analysis of progressives and habituals, given that imperfective forms in many languages display syncretism for these two meanings. Given that Wolof *di* can also be used for both progressive and habitual readings, a unified analysis is desirable.

The main innovation of Ferreira's analysis is the idea that the basic machinery of Portner's analysis applies to both progressives and habituals, and that progressives apply to singular events, while habituals apply to plural events. Ferreira proposes the following SG and PL operators that apply above VP but below Asp. The VP is taken do denote a set of events, including both atomic and pluralities of events. As shown in (36), SG applies to a VP denotation and returns a set of atomic events, while PL applies to a VP denotation and returns a set of plural events.

(36) a. SG($[\![$VP$]\!]^w$) = $\{e_1, e_2, e_3, \ldots\}$
b. PL($[\![$VP$]\!]^w$) = $\{e_1 \oplus e_2, e_2 \oplus e_3, e_1 \oplus e_3, e_1 \oplus e_2 \oplus e_3, \ldots\}$ (Ferreira 2016: 358)

The run time of a plural event is derived from a plurality of time intervals, as in (37). Crucial to the idea of habituals as plural events is the temporal inclusion relation defined in (38). This ensures that an interval i may be included in $\tau(e \oplus e')$, even if i is disjoint from both $\tau(e)$ and $\tau(e')$. This accounts for the intuition that there may be intervals within the run time of a habit where a particular event that instantiates the habit is not taking place; i.e., Fatou need not be drawing a circle at the utterance time of (20) for the sentence describing the habit to be true. This is illustrated in (39).

(37) $\tau(e \oplus e') = \tau(e) \oplus \tau(e')$

(38) Inclusion:
An interval i is included ($\subseteq$) in an interval i' iff the left boundary of i' precedes the left boundary of i and the right boundary of i precedes the right boundary of i'.

(Ferreira 2016: 361)

(39) ——$[_{i1}$——$]$—$[_{i3}$——$]$—$[_{i2}$——$]$—> $i_3 \subseteq i_1 \oplus i_2$

Putting this all together, the truth conditions of the habitual sentence in (20), repeated below as (40a), can be modeled as in (40b). The sentence is true if there is a plural event (i.e., a habit) of circle drawing by Fatou that is ongoing at the reference time, and that habit continues into the future in the inertia worlds because the reference time is a non-final sub-interval of the run-time of the habit in the inertia worlds. The truth conditions in (40b) together with the inclusion condition of (38)-(39) ensure that there need not be an event of Fatou drawing a circle ongoing at the reference time.

(40) a. (Altine bu nekk,) Faatu daf-a=∅ di (>dafay) rëdd ay wërëngërël.
Monday C$_{\text{REL}}$ be, Fatou do-C=3SG.S DI draw INDF.PL circle
'(Every Monday,) Fatou draws circles.'

b. ⟦*Faatu dafay rëdd ay wërëngërël*⟧
$= 1$ iff $\exists e \exists t[t \subseteq \tau(e)$ & $\forall w' \in \textsc{best}(\textsc{circ}, \textsc{ni}, e)[\exists e' \exists t'[t \subset_{nf} t'$
& $t' = \tau(e')$ & $\textsc{pl}(\mathbf{draw}(f, c, e', w'))]]]$

For completeness, we update the analysis of the progressive reading in (35b) to include the singular VP operator in (36). The truth conditions of (41) are identical to those in (40b), except for the PL operator swapped for the SG operator, and the identity of the subject.

(41) ⟦*damay rëdd wërëngërël*⟧
$= 1$ iff $\exists e \exists t[t \subseteq \tau(e)$ & $\forall w' \in \textsc{best}(\textsc{circ}, \textsc{ni}, e)[\exists e' \exists t'[t \subset_{nf} t'$ & $t' = \tau(e')$
& $\textsc{sg}(\mathbf{draw}(sp, c, e', w'))]]]$

To summarize, the progressive and habitual readings associated with Wolof *di* can be given a unified analysis by adopting the Portner/Ferreira analysis of progressives and habituals. In both cases, there is an event that is ongoing at the reference time, which continues and culminates in all the inertia worlds. In the next section, we turn to discuss the future readings of *di*, which we also aim to capture by extending the basic analysis presented here.

5 Flavours of the future

As shown in Section 3, *di* can be used for future readings, in addition to progressive and habitual readings. Recall as well that when *di* is in a higher position in the syntax, only a future reading is available. In this section, we discuss a further quirk in the puzzle, namely that within future readings, the position of *di* has an effect on the "flavour" of the future reading. Our consultants are very clear that the following sentences are not interchangeable across contexts (i.e., (42) vs. (43) and (44) vs. (45)), and the judgments are robust across speakers.

(42) CONTEXT: Fatou is a queen, so her son will automatically become the king one day.
Faatu xam-na-∅ ni doom-om, ba mu juddoo, daf-a=∅ **di** (>dafay)
Fatou know-C=3SG.S that child-POSS.3SG when 3SG.S be.born do-C=3SG.S DI
nekk buur.
be king
'Fatou knows that her son, when he is born, is going to be king.' (low *di*)

(43) CONTEXT: Fatou is poor, but she had a dream that she will have a son who will one day become king.
Faatu xam-na=∅ ni doom-om, ba mu juddoo, **di**-na=∅ nekk buur
Fatou know-C=3SG.S that child-POSS.3SG when 3SG.S be.born DI-C=3SG.S be king
'Fatou knows that her son, when he is born, is going to be king.' (high *di*)

(44) CONTEXT: The addressee is already engaged to a rich woman, so the marriage is certain.
Yow da-nga **di** (> dangay) takk jigeen ju am xaalis.
2SG do-C.2SG DI be.attached woman C$_{\text{REL}}$ have money
'You will marry a rich woman.' (low *di*)

(45) CONTEXT: A fortune teller predicts:

Yow	**di**-nga	takk	jigeen	ju	am	xaalis.
2SG	DI-C.2SG	be.attached	woman	C_{REL}	have	money

'You will marry a rich woman.' (high *di*)

The intuition about these contrasts that we wish to capture is the following. Low *di* futures are used to make a prediction based on certain facts or circumstances that hold at the speech time. For instance in (42), relevant facts are that Fatou is the queen, combined with general rules of royal succession. In (44), the relevant fact is that the addressee is already engaged to be married to a rich woman. Meanwhile, high *di* futures are predictions not based on any particular circumstances at the speech time, but are just pure predictions. For instance in (43), Fatou is not the queen; she predicts her son will be king based on a dream she had. Similarly in (45), the fortune teller simply has knowledge of the future and makes the prediction, independent of any current circumstances.

To preview our proposal, we argue that future readings of low *di* employ a circumstantial modal base, whereas future readings of high *di* use an epistemic modal base.

5.1 Low *di*: Circumstantial futures

Our proposal for low *di* is that it expresses a future reading with a circumstantial modal base. We aim to extend the analysis we have developed so far for progressive and habitual readings, which also make use of a circumstantial modal base, to account for future readings of low *di* as well. However, our semantics so far requires that an event be ongoing at the reference time, which seems at odds with a reading where an event takes place wholly in the future. We propose that the account can be maintained by taking into account preparatory stages of events. To give some background to this idea, we follow the discussion of Grano (2011) in his analysis of English *try*.

Grano (2011), building on ideas of Sharvit (2003), provides an analysis of English *try*, whereby TRY(e, P, a, w) is true in a world w if and only if an agent a is engaged in an event e that has set into motion preconditions that develop into event e' of type P in worlds w'. What counts as a precondition that develops into an event e' of type P is context-sensitive, but could include mental action or effort to bring e' about. TRY(e, P, a, w) is also satisfied if a is actively engaged in an event of type P at the reference time. This analysis of *try* thus bears important temporal similarities to progressives and habituals as presented in the previous section: an event is ongoing at the reference time, which further develops beyond the reference time into an event of the sort named by the predicate P. For progressives and habituals, the inner stage of the event is ongoing at the reference time, whose endpoint occurs in the future of the reference time in the inertia worlds. For futures, we can say that it is a preparatory stage that is ongoing at the reference time, which places the inner stage (i.e., the main action) of the event wholly in the future of the reference time. This distinction is schematized in (46).

(46) ——————————I——————————I——————————

PREPARATORY INNER STAGE ENDPOINT RESULT STATE

[- - - - - - FUT - - - - - -] [- - - - - - PROG - - - - - -]

[- - - - - - - - - - - - - - - low *di* - - - - - - - - - - - - - -]

Under this view, there are similarities between the future reading of low *di* and future readings of the present tense in English, also known as 'futurates' (Copley, 2002, 2008; Rullmann et al., 2023). The English futurate construction is formed either with the simple present or a present progressive form to convey a future meaning, as illustrated in (47).

(47) The Red Sox {play/are playing} the Yankees tomorrow.

Copley (2002, 2008) argues that futurates involve a covert modal operator PLAN, which requires a metaphysical modal base with a stereotypical ordering source. PLAN furthermore presupposes that there is a director (agent) that directs the prejacent proposition p at the reference time. Roughly, this means that the director has the ability to bring about that p, and is committed to bringing about that p. The director presupposition aims to account for why events that cannot normally be planned sound odd with futurates, such as (48).[5] Rullmann et al. (2023) argue that plain futurates are only acceptable where a salient schedule of events holds at the reference time. They also predict the plain futurate in (48) to be unacceptable, since winning or losing a baseball game is not part of a schedule (barring cheating contexts).[6]

(48) #The Red Sox {defeat/are defeating} the Yankees tomorrow.

Under both views, the reference time of a futurate is the present, whereby some pre-condition for a future event (i.e., a plan or schedule) holds at the present. This is similar to the case of *try* as argued by Grano, where at least a preparatory stage of an event must be ongoing at the reference time.

Carrying these ideas over to our case of Wolof *di*, we suggest that the analysis that we have developed so far can be extended to account for the future reading of low *di*; namely, a circumstantial modal base and non-interruption ordering source can account for this reading as well. Specifically, we propose that a preparatory stage consisting of a plan, schedule or mental action ongoing at the reference time can be enough to satisfy the event-in-progress part of the semantics. When a preparatory stage of the event is ongoing at the reference time, then the main action or inner stage of the event is located in the future of the reference time. Not only is the preparatory stage part of the circumstances that must hold in the best worlds, but also the dispositions and abilities of the event participants (e.g., that an agent has the ability to make an event come about or has the authority to make and enforce a schedule).[7]

Our analysis of the sentence (22), repeated as (49), is given in (50). The truth conditions are essentially identical to the event in progress reading, but in this case the event e that is ongoing at the reference time is a preparatory event, in this case a planning event, and e' is a temporal super-event that includes the planning event and the inner stage of the event, i.e., the event of actually building the house, which takes place fully in the future of the reference time.

(49) [CONTEXT: I see Mbaye walking around the town with an architect, buying building materials, etc., and I ask our mutual friend what Mbaye is up to, and he tells me:]
Mbaye daf-a=∅ di (> dafay) tabax kër.
Mbaye do-C=3SG.S DI build house
'Mbaye is going to build a house.'

(50) $\exists e \exists t[t \subseteq \tau(e)$ & $\forall w' \in \text{BEST}(\text{CIRC}, \text{NI}, e)[\exists e' \exists t'[t \subset_{nf} t'$ & $t' = \tau(e')$
& $\text{SG}(\textbf{build.a.house}(m, e', w'))]]]$

To sum up so far, we have argued that the progressive, habitual and future readings for (low) *di* can be accounted for in a unified way by adopting the Portner/Ferreira analysis of progressives and habituals, and extending that analysis to circumstantial future readings. However, we submit that not all future uses of *di* can be analyzed in this way. Specifically, there are cases where there is no obvious planning event or other preparatory stage of a future event, and where a claim about the future is a pure prediction based on speculation. We turn to such cases next.

[5] In later work, Copley (2014) offers an analysis of futurates involving causal chains.

[6] Rullmann et al. note that progressive futurates can also describe a schedule, but also have a wider applicability.

[7] Portner (1998) argues that dispositions and abilities need to be taken into account for the progressives, and Ferreira (2016) shows argues the same for habituals.

5.2 Another modal base for *di*

As we have seen, there are two "flavours" of future readings with *di*; a minimal pair is shown in (51)-(52), repeated from above. The example in (51) represents the future reading of low *di*, which we analyzed in the previous subsection as an extension of the progressive and habitual readings of low *di*, i.e., with a circumstantial modal base. However, the future reading in (52) with high *di* does not seem amenable to such an analysis, since there does not seem to be a plausible plan, schedule or other preparatory stage ongoing at the reference time. This is the case for the other uses of high *di* that we have seen.

(51) CONTEXT: The hearer is already engaged to a rich woman, so the marriage is certain.
Yow da-nga di (> dangay) takk jigeen ju am xaalis.
2SG.S do-C.2SG.S DI be.attached woman C_{REL} have money
'You will marry a rich woman.' (low *di*)

(52) CONTEXT: A fortune teller predicts:
Yow di-nga takk jigeen ju am xaalis.
2SG.S DI-C.2SG.S be.attached woman C_{REL} have money
'You will marry a rich woman.' (high *di*)

We will defer until the next subsection the explanation for why these particular readings are dependent on the syntactic position of *di*. For now, our aim is to characterize the future reading in sentences like (52).

We argue that the future readings of high *di* are derived from an epistemic modal base, rather than a circumstantial one. As we have already explained, under a Kratzerian view on modals (Kratzer, 1981, 2012), different modal flavours for one and the same modal is due not to ambiguity, but to the availability of different modal bases given in the context. Thus, the different flavours of the future uses of *di* can be accounted for by saying that *di* is compatible with different modal bases, so that the flavours of future *di* are akin to variability of the modal flavour found in many other modals.[8]

As a first pass at capturing the truth conditions of future sentences with high *di*, we propose that high *di* makes use of an epistemic modal base instead of a stereotypical one. Applying this idea to the sentence (26), repeated here as (53), we sketch its truth conditions in (54). The sentence is true iff in all the best epistemically accessible worlds w' at the reference time t, there is an event of Moussa coming whose run time is t', which is located in the future of the reference time.

(53) Musaa di-na=∅ ñëw.
Moussa DI-C=3SG.S come
'Moussa will/is going to come.'

(54) $[\![\textit{Musaa dina ñëw}]\!] = 1$ iff $\exists t[\forall w' \in \textsc{best}(\textsc{epist}, \textsc{ster}, t)[\exists e \exists t'[t' > t \,\&\, t' \circ \tau(e)$
$\& \textbf{ come}(m, e, w')]]]$

Taking stock so far, the event-in-progress, habitual and future readings of low *di* are derived from a circumstantial modal base, whereas the future readings of high *di* are derived from an epistemic modal base. This analysis captures the difference in flavour that future statements with *di* can have, namely we derive the different flavours of the future using different types of modal bases. The circumstantial future is based on the circumstances surrounding the event of the verb phrase (e.g., a plan or preparatory stage of the event that continues into the future), whereas the

[8]The idea that modal operators other than modal auxiliaries can also be compatible with a variety of modal flavours is also present in the work of Arregui et al. (2014), who analyze variability in modal flavours available for imperfectives cross-linguistically (including future-oriented imperfectives in some Slavic languages).

epistemic future is simply a pure prediction or speculation ("for all I know..."), not necessarily grounded in any circumstances that hold at the reference time.

Furthermore, this data and our analysis of it contribute to an ongoing debate in the literature about the flavour of modal future statements. While certain authors (e.g., Condoravdi 2002; Copley 2002, 2008; Kaufmann 2005) use a metaphysical modal base with inertial or bouletic ordering sources, others (e.g., Tonhauser 2011; Giannakidou & Mari 2018) claim the future is epistemic. We argue that both routes to future readings are possible, as evidenced by the different future flavours that Wolof *di* can give rise to.

The sketch of the analysis here for high *di* will be revised in the next section, since the temporal/aspectual part of the analysis in (54) is quite different from what was proposed for low *di*, and we aim to offer a unified analysis. Notably, the event-in-progress and the non-final temporal interval aspects of the analysis are missing in (54). First, though, we wish to explain why the modal flavours of *di* are restricted based on its syntactic position.

5.3 Modal variability and syntactic height

The systematic mapping between modal flavour and syntactic position is reminiscent of the analysis of modal flavours by Hacquard (2006, 2010), whereby the syntactic position of a modal determines the type of modal base that it associates with. Specifically, a circumstantial modal base is derived for modals located below tense, and an epistemic modal base is derived for modals located above tense. Under Hacquard's analysis, the differences in flavour are derived from the availability of different event anchors for generating the modal base, and not on multiple lexical entries for the modals specifying their flavours, following Kratzer's (1981; 2012) insights. Low-scoping modals (below T) only have access to a circumstantial modal base, while high-scoping modals (above T) only have access to an epistemic modal base. The analysis thus invokes a single lexical entry for modal auxiliaries, but where the choice of modal base is syntactically determined.

Under Hacquard's analysis, the modal anchor is always an *evaluation event* (rather than an *evaluation world* on more classical analyses), but the identity of the event is different in the low versus high positions. For low-scoping modals, the evaluation event is the event that is bound by the aspect head – the event that the sentence is "about". The possibilities generated from this event relate to the circumstances of that event, including its participants (e.g., the agent) and their dispositions at the time the event is ongoing. In effect, these are the circumstances associated with a circumstantial modal base. For high-scoping modals, the evaluation event is the speech event of uttering the sentence.[9] The possibilities generated from this event also relate to the circumstances of that event: its participants (e.g., the speaker) and their dispositions at the time the event is ongoing (i.e., the speech time). As Hacquard argues, a speech event, being a type of attitude event, is associated with propositional content, namely an information state that an epistemic modal quantifies over.

Although we have been focusing on clauses without any morphological tense, we do find corroborating evidence from optional past tense (Bochnak & Martinović, 2017) for the view that the distinct structural positions of *di* determine its possible interpretations. If the optional tense morpheme *-oon* is present in a clause without *di*, it suffixes onto the main verb; if the verb moves to C, the tense morpheme is carried with it, as in (55).

(55) Xale yi lekk-**oon**-na=ñu jën.
child the.PL eat-PST=C=3PL.S fish
'The children ate fish.'

[9]In complements of attitude verbs, it is the attitude event (e.g., believing, thinking, saying, etc.). In that sense, high-scoping modals are always related to an attitude event: the attitude of a matrix clause is one of saying.

When *oon* affixes onto low *di*, the result is a past progressive reading, as in (56). However, as shown in (57), the past progressive combination of *d(i)-oon* cannot appear in C. That is, the past progressive reading is restricted to sentences where *di* remains low.[10] This provides further evidence that the circumstantial readings are not available for high *di*.

(56) Xale yi da=ñu d(i)-oon lekk jën.
child the.PL do.C=3PL.S DI-PST eat fish
'The children were eating fish.'

(57) *Xale yi d(i)-oon-na=ñu lekk jën.
child the.PL DI-PST-C=3PL.S eat fish
intended: 'The children were eating fish.'

Thus, we find that the choice of modal base for *di* depends on its syntactic position. A circumstantial modal base for low *di* is already part of our analysis: in following Portner (1998), we incorporate a modal base generated by the event described by the verb phrase: the circumstances surrounding that event and its participants derive the modal interpretation of the progressive, habitual, and circumstantial futures. To extend this analysis to high *di*, we follow Hacquard in positing an event-based modal semantics for epistemic modality as well.

As we have said, under Hacquard's account, modals are relativized to an evaluation event, and for epistemic modals, the evaluation event is the speech event. The speech event is associated with an information state. Hacquard then identifies an epistemic modal base $\text{EPIS}(e)$ with the content of the speech event, $\text{CON}(e)$. The set of worlds compatible with the content of the speech event are then the speaker's doxastic alternatives. This is summarized in (58).

(58) a. $\text{EPIS}(e) = \text{CON}(e)$
b. $\cap\text{CON}(e) = \text{DOX}(\iota x : \text{Holder}(x, e), w)$
c. $\iota x : \text{Holder}(x, e) = \text{speaker}(e)$ (adapted from Hacquard 2010)

In sum, an event-based semantics for modals can be used to understand why there is a relation between the syntactic position of a modal and the range of flavours that can be expressed in that position. In the analysis that we develop in the next section, *di* will be a modal that receives a circumstantial modal base in its low position and an epistemic modal base in its high position.

6 Deriving the readings of high and low *di*

A challenge for unifying the high and low readings of *di* lies in the temporal/aspectual differences between these two uses. For low *di*, there is always an event that is ongoing at the reference time that develops, in all the best accessible worlds, into an event of the relevant kind. The nature of that event gives rise to the different interpretations of low *di*: if it is a singular event that is in progress, the progressive reading obtains; if it is a habit *qua* plural event, the habitual reading obtains; and

[10] Examples as in (57) have been reported as grammatical in, e.g., Torrence 2003; they are ungrammatical for all our speakers, who note that the intended form is probably the one in (i).

(i) Xale yi daan-na=ñu lekk jën.
child the.PL PST.HABIT-C=3PL.S eat fish
'The children used to eat fish.'

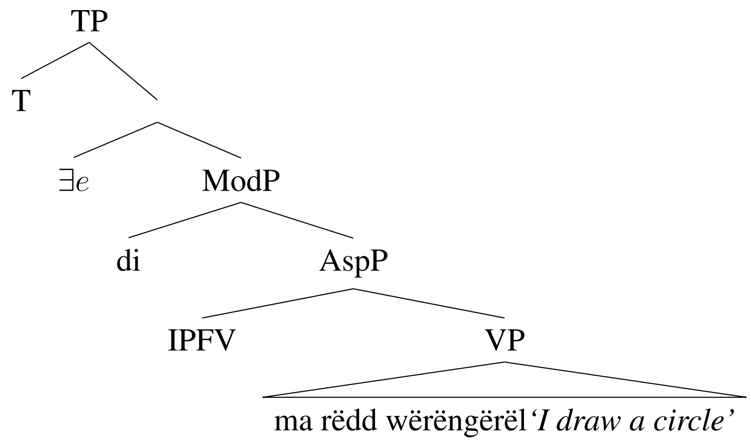
Daan is a past habitual auxiliary, which is not clearly decomposable into *di* and another morpheme, so we take it to be monomorphemic. For many speakers, the past progressive *di-oon* and the past habitual *daan* have neutralized into one form, namely *daan*. For more on this, see Martinović 2019.

if it is a preparatory stage of an event, the future reading obtains. For high *di*, there is no such event ongoing at the reference time other than the speech event itself. But there is intuitively no sense in which the speech event develops into an event of the sort that is described by the VP. The VP event simply takes place at a time in the future of the speech time. Thus, while the modal contribution of *di* remains constant across its uses (assuming a Hacquard-style analysis of deriving flavour differences syntactic position, as we do), the temporal/aspectual profile of high vs. low di cannot be so easily unified.

Our proposal is to decompose the modal and aspectual components that we have observed in sentences containing *di*. We will treat *di* essentially as a necessity modal that is compatible with both circumstantial and epistemic modal bases. The modality of *di* is event-relative, and so in this respect we follow Hacquard's analysis quite closely. The aspectual semantics is contributed by other operators in the clauses in which *di* occurs.

Let us begin with *di* in its low position, which can take on progressive, habitual, and circumstantial future readings. We propose that these readings come about through a combination of *di* and a covert imperfective aspect that *di* scopes over. We will illustrate the proposal using the sentence (18), repeated here as (59), which has a progressive reading. The structure we propose for (59) is shown in (60).

(59) CONTEXT: I am standing in front of a wall with a bucket of paint and I just put a brush to the wall and started drawing something. Someone walks into the room and asks *What are you doing?* I respond:
Da=ma di (> damay) rëdd wërëngërël.
do.C=1SG.S DI draw circle
'I am drawing a circle.'

(60)


In this configuration, *di* licenses and must co-occur with a covert imperfective aspect. Our semantics for the imperfective aspect in (61) is fairly standard in placing the reference time of the clause within the event time (e.g., Kratzer 1998), but with two modifications. The first is the specification that the reference time cannot be a final sub-interval of the event time ($t \subset_{nf} \tau(e)$). The second is that the event variable does not get existentially bound within IPFV, but remains abstracted over. In this respect, we are following the precedent of Cable (2013), where aspect encodes the relation between the eventuality time and the reference time, but the existential force is external to the aspect head itself. The advantage for us is that this move allows the event variable to be passed up to *di*. Assuming (62a) as the denotation of the VP in (60), the result of adding IPFV is shown in (62b).

(61) $[\![$ IPFV $]\!] = \lambda P_{\langle v,st\rangle} \lambda e \lambda t \lambda w . t \subset_{nf} \tau(e)$ & $P(e)(w)$

(62) a. $[\![$ VP $]\!] = \lambda e \lambda w.$SG(**draw.circle**$(e)$ & **Ag**$(e) =$ *sp* & e in w)

b. $[\![$ IMPF(VP) $]\!] = \lambda e \lambda t \lambda w. t \subset_{nf} \tau(e)$ & SG(**draw.circle**(e) & **Ag**$(e) = sp$ & e in w)

This in turn is the input to the modal *di*, which has the following requirements. Event e overlaps with the reference time, and in all the best worlds w' where the circumstances of e hold and continue stereotypically, there is an event e' of type $\mathcal{P}$ in w'. This is spelled out in (63), where we collapse the non-interruption ordering source of low *di* into the notion of stereotypicality. The result of applying this to the denotation of AspP is shown in (64), along with an informal paraphrase of the truth conditions that are derived.

(63) $[\![\text{di}]\!] = \lambda \mathcal{P}_{\langle v, \langle i, st \rangle \rangle} \lambda e \lambda t \lambda w. \tau(e) \circ t \;\&\; \forall w'[w' \in \text{BEST}(\cap f(e), \text{STEREO}, t, w) \rightarrow \exists e'[\mathcal{P}(e')(t)(w')]]$

(64) $[\![\text{di(IPFV(VP))}]\!] = \lambda e \lambda t \lambda w. \tau(e) \circ t \;\&\; \forall w'[w' \in \text{BEST}(\cap f(e), \text{STEREO}, t, w)$
$\rightarrow \exists e'[t \subset_{nf} \tau(e')$ & SG(**draw.circle**(e') & Ag$(e') = sp$ & e' in w')]]

Informal paraphrase: "e overlaps with t, and in all inertia worlds w' based on the circumstances of e, there is an event e' which is a circle-drawing event by the speaker, where t is a non-final sub-interval of the run-time of e' " (i.e., e is ongoing at t, and culminates as e' in all inertia worlds w')

The lambda-bound event variable is then existentially bound above *di*, and the reference time variable is taken care of by tense. We will be more specific about the role of tense when we discuss the derivation of high *di* below. For exposition, we will just assume for now that the reference time variable is existentially bound by the T head, and we derive the final truth conditions for low *di* (on the progressive reading) are given in (65).

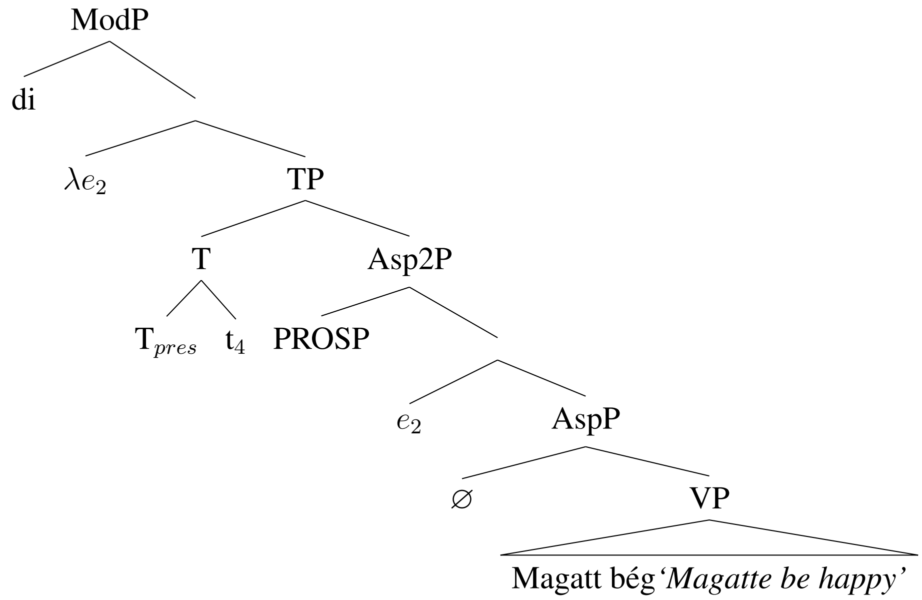
(65) $[\![(59)]\!] = \lambda w. \exists e \exists t[\tau(e) \circ t \;\&\; \forall w'[w' \in \text{BEST}(\cap f(e), \text{STEREO}, t, w)$
$\rightarrow \exists e'[t \subset_{nf} \tau(e')$ & SG(**draw.circle**(e') & Ag$(e') = sp$ & e' in w')]]]

The same setup will derive the habitual and circumstantial readings of low-*di*, given what we have already said. The habitual reading is derived by swapping PL for SG at the VP level. The circumstantial future reading obtains when the event e that feeds aspect and *di* includes a preparatory stage that is ongoing at the reference time.

For high-*di*, we assume the structure in (67), where *di* is located above T. We illustrate our analysis using the sentence in (66), repeated from (25).

(66) CONTEXT: Mbaye is making his wife Magatte's favorite dish for lunch and I comment to him:
Magatt di-na=∅ bég.
Magatte DI-C=3SG.S happy
'Magatte is going to be happy.'

(67)



In this setup, *di* licenses and co-occurs with a covert prospective aspect, which introduces future-shifting for high-*di*. We assume, as is standard, that ordering aspects are located between viewpoint aspect and tense. The semantics of the prospective aspect is as in (68).

(68) $[\![\text{PROSP}]\!] = \lambda \mathscr{P}_{\langle i,st \rangle} \lambda t \lambda w. \exists t'[t' > t \ \& \ \mathscr{P}(t')(w)]$

In Asp, we posit a null empty aspect, whose contribution is to introduce a temporal variable that is identified with the event run time of the VP, as shown in (69) (cf. Matthewson 2012).

(69) $[\![\varnothing \text{ (empty aspect)}]\!] = \lambda P \lambda e \lambda t \lambda w. t = \tau(e) \ \& \ P(e)(w)$

As with the imperfective in (61), the event variable does not get bound by aspect, but gets filled in by an event pronoun external to AspP. With these pieces in hand, the semantic composition up to the prospective aspect is shown in (70).

(70) a. $[\![\text{VP}]\!] = \lambda e \lambda w. \textbf{be.happy}(e) \ \& \ \text{Hold}(e) = \text{M} \ \& \ e \text{ in } w$

b. $[\![\varnothing(\text{VP})]\!] = \lambda e \lambda t \lambda w. t = \tau(e) \ \& \ \textbf{be.happy}(e) \ \& \ \text{Hold}(e) = \text{M} \ \& \ e \text{ in } w$

c. $[\![\varnothing(\text{VP})(e_2)]\!] = \lambda t \lambda w. t = \tau(e_2) \ \& \ \textbf{be.happy}(e_2) \ \& \ \text{Hold}(e_2) = \text{M} \ \& \ e_2 \text{ in } w$

d. $[\![\text{Asp2P}]\!] = \lambda t \lambda w. \exists t'[t' > t \ \& \ t' = \tau(e_2) \ \& \ \textbf{be.happy}(e_2) \ \& \ \text{Hold}(e_2) = \text{M} \ \& \ e_2 \text{ in } w]$

As for tense, Wolof has previously been analyzed as an optional tense language, where clauses with no overt tense contain an underspecified temporal pronoun in T (Bochnak & Martinović, 2017). This view is not essential for our purposes here, and we can just as well make use of a null non-future tense (Matthewson, 2006) or even distinct null present and past tenses. For perspicuity, we posit the present tense in (71a), which presupposes that t is identified with the speech time t_c.

(71) a. $[\![\text{T}_{pres}]\!] = \lambda t \lambda \mathscr{P}_{\langle i,st \rangle} \lambda t'' \lambda w. \mathscr{P}(t)(w) \ \& \ \partial(t = t'')$

b. $[\![\text{TP}]\!] = \lambda t'' \lambda w. \exists t'[t' > t'' \ \& \ t' = \tau(e_2) \ \& \ \textbf{be.happy}(e_2) \ \& \ \text{Hold}(e_2) = \text{M} \ \& \ e_2 \text{ in } w] \ \& \ \partial(t_4 = t'')$

c. $[\![\text{node above TP}]\!] = \lambda e_2 \lambda t'' \lambda w. \exists t'[t' > t'' \ \& \ t' = \tau(e_2) \ \& \ \textbf{be.happy}(e_2) \ \& \ \text{Hold}(e_2) = \text{M} \ \& \ e_2 \text{ in } w] \ \& \ \partial(t_4 = t'')$

d. $[\![\text{ModP}]\!] = \lambda e \lambda t \lambda w. \tau(e) \circ t \ \& \ \forall w'[w' \in \text{BEST}(\cap f(e), \textsc{stereo}, t, w) \rightarrow \exists e' \exists t'[t' > t \ \& \ t' = \tau(e') \ \& \ \textbf{be.happy}(e') \ \& \ \text{Hold}(e') = \text{M} \ \& \ e' \text{ in } w']] \ \& \ \partial(t_4 = t)$

At this stage, the event and time variables get filled in by the speech event and time. Crucially, the speech event will yield an epistemic modal base for $\cap f(e)$, following Hacquard's analysis in (58). We thus derive that high-*di* has an epistemic reading only, and gives rise to a future reading by (obligatorily) co-occurring with a null prospective aspect.

In sum, our analysis derives the correlation between modal flavour and syntactic position by adapting Hacquard's event-based analysis for modals. In our analysis, *di* on its own is a modal element only, and the aspectual effects (imperfective and prospective interpretations) are factored out into null aspectual heads, which must be licensed by a c-commanding *di*. A question that arises from this analysis is what controls the distribution of the null aspectual operators that we have proposed. Considering first the epistemic future readings with high *di*, we suggest that the prospective is required since its absence would give rise to an identical reading to what would be derived from the modal verb *war* 'must', as in (72).

(72) CONTEXT: Moussa always comes to Idy's for afternoon tea. Even if he is not feeling well, he never misses a day. This isn't an obligation for Moussa, he just likes having tea with Idy every day. It is tea-time, so...
Musaa **war** na=∅ nekk ci Idy.
Moussa must C=3SG.S be at Idy
'Moussa must be at Idy's.'

On the flip side, it remains mysterious why the covert prospective aspect could not be licensed by *war* to derive future-oriented epistemic readings. Since a present-oriented epistemic reading can independently be expressed through *war*, we suggest that *di* and *war* have developed separate niches in the language, whereby epistemic *di* must have future orientation, but epistemic *war* does not. The nature of this conventionalization of readings remains a question for future research. As for the wider distribution of the covert aspectual operators, we suggest that these may be licensed by other verbs as well, which we discuss in section 7.2 below.

7 Discussion and extensions

In this section, we discuss the differences between *di* and other modal verbs in Wolof, and we sketch how our analysis can be extended to cases where *di* occurs in the complement of control predicates.

7.1 Other modal verbs

Wolof has two modal verbs, necessity *war* 'must' and possibility *mën* 'can'. These behave differently from *di* both syntactically and semantically. Syntactically, *war* and *mën* take an infinitival complement. When the matrix and the infinitival verbs are adjacent, a linker element *a* occurs between them; Torrence (2012) considers it an infinitival complementizer, but Martinović (2024) argues against this view, showing that infinitival complements in subject control constructions are smaller than a CP. She proposes that *a* is just a linker vowel, similar to linkers in Greek and Slavic compounds (Nespor and Ralli 1996; Gouskova 2010). Like other lexical verbs in Wolof, *war* and *mën* can appear in both low and high positions within the clause. (73)-(74) show these verbs in the low position. In these contexts, they receive an epistemic reading.

(73) CONTEXT: You have a headache that isn't going away, so you see the doctor. All your tests come back clean, so the doctor says:
Lii daf-a=∅ **war a** nekk coono wala jaaxle.
this DO-C=3SG.S must *a* be fatigue or anxiety

'This must be fatigue or anxiety.'

(74) CONTEXT: You are feeling tired a lot. The doctor does various tests and finds that you are anemic. She will do more tests to exclude other illnesses, but in the meantime she gives you some iron pills and says:
Lii daf-a=∅ **mën a** nekk ñakk-deret.
this DO-C=3SG.S can *a* be lack-blood
'This could be anemia.'

Like other verbs in the language, *war* and *mën* raise to C in structures with the sentence particle *na*. This is shown for *war* in (75) and for *mën* in (76).

(75) CONTEXT: Moussa always comes to Idy's for afternoon tea. Even if he is not feeling well, he never misses a day. This isn't an obligation for Moussa, he just likes having tea with Idy every day. It is tea-time, so...
Musaa **war** na=∅ nekk ci Idy.
Moussa must C=3SG.S be at Idy
'Moussa must be at Idy's.'

(76) CONTEXT: Every day at 4 pm, Moussa is either having tea at Idy's, or he is taking a walk. It is 4 pm, so...
Musaa **mën** na=∅ nekk ci Idy.
Moussa can C=3SG.S be at Idy
'Moussa could be at Idy's.'

Like the English modal auxiliaries *must* and *can* as well as Wolof *di*, the modal verbs *war* and *mën* are compatible with both epistemic and circumstantial interpretations. What is noteworthy about *war* and *mën* in comparison with *di* is that the position of *war* and *mën* has no effect on the available flavours for these modals. As was shown in (73)-(76), *war* and *mën* can have an epistemic interpretation in both the low and high positions. (77) and (78) show circumstantial readings for these verbs in the low position, while (79) and (80) show circumstantial readings in the high position.

(77) CONTEXT: The hospital rules say...
Ñii seet si malaad yi da=ñu **a** (>dañoo) **war a** dem balaa fukk-i
DEM see at sick.one the.PL do.C=3PL.S *a* must *a* go before ten-GEN.PL
waxtu ak juróom ñett.
hour and five three
'Those visiting the sick must leave before 18:00 hours.'

(78) Da=ma **mën a** daw.
do.C=1SG.S can *a* run
'I can really run./I am a great runner.'

(79) CONTEXT: You have diabetes but mostly eat a diet rich in carbohydrates. Your doctor imposes a strict diet and says:
War nga lekk lujum.
must C.2SG.S eat vegetables
'You must eat vegetables.'

(80) CONTEXT: You have a job interview, and you've arrived carrying a briefcase. The receptionist tells you:

Mën nga=ko=fi wàcc, mën nga=ko tamit yobbale.
can C.2SG.S=3SG.O=LOC descend can C.2SG.S=3SG.O also carry.with
'You can leave it here, you can also carry it with you.'

Clearly, *war* and *mën* contrast with *di* in terms of the availability of modal flavours in different positions, with syntactic position showing no correlation with the availability of modal flavours for *war* and *mën*. We propose that *mën* and *war* are not event-relative like *di* – they do not depend on an event argument to deliver (different) modal flavours; they just need some modal base available from the context, along the lines of a more traditional Kratzerian analysis. They can be analyzed along the lines of (81).

(81) a. $[\![\textit{war}]\!] = \lambda f_{\langle s,\langle st,t\rangle\rangle} \lambda g_{\langle s,\langle st,t\rangle\rangle} \lambda p_{\langle s,t\rangle} \lambda w_s . \forall w' \in \text{BEST}(f,g,w) : p(w') = 1$
b. $[\![\textit{mën}]\!] = \lambda f_{\langle s,\langle st,t\rangle\rangle} \lambda g_{\langle s,\langle st,t\rangle\rangle} \lambda p_{\langle s,t\rangle} \lambda w_s . \exists w' \in \text{BEST}(f,g,w) : p(w') = 1$

Syntactically, *di* is an auxiliary which can be generated in two different positions, and which receives two different interpretations based on the event variables available in those positions. Meanwhile, *war* and *mën* are like regular lexical verbs, and can head-move to C like other verbs, but movement to a higher position does not affect the interpretation. This shows that the event-relativity of modals is a lexical fact, and event-relative and non-event-relative modals can co-exist in the same language.

7.2 Embedded *di* with control predicates

Restructuring verbs like *try* are generally assumed to take small complements; Martinović (2024) argues for Wolof that they take vPs based on clitic climbing patterns. Namely, clitic climbing is obligatory in Wolof with typical restructuring verbs (modals, aspectual verbs, implicatives, *try*), but optional wtih typically non-restructuring verbs, such as desideratives. The obligatoriness of clitic climbing is illustrated with the implicative verb *sàggane* 'neglect' in (82), and the optionality with the verb *fasyéene* 'decide' in (83) (the optionality of clitic climbing is represented with curly brackets, which have a disjunctive interpretation vis-a-vis clitic position).

(82) a. Sàggane na=a=**leen** [indil tàngal __].
neglect C=1SG.S=3PL.O bring candy __
'I neglected to bring them candy.'
b. *Sàggane na=a [indil=**leen** tàngal __]. (Martinović, 2024, p.1596)

(83) Faatu fasyéene na=∅={**ko**} [togg={**ko**}].
Fatou decide C=3SG.S=3SG.O cook=3SG.O
'Fatou decided to cook it.' (Martinović, 2024, p.1596)

Martinović (2024) argues that clitic climbing is obligatory out of infinitives that do not contain the inflectional layer (i.e., that are vP-sized), because clitics must be licensed by an inflectional head. She argues that complements of typically non-restructuring verbs are TP-sized, and therefore clitics may remain inside the infinitival clause (for additional arguments and a detailed analysis of Wolof infinitives and cliticization, see Martinović 2024, to appear). She concludes this because *di* optionally occurs in the complements of restructuring verbs, and when it does, clitic climbing becomes optional, just as it is out of TP-sized infinitives, as illustrated in (84), with a modal verb where clitic climbing is otherwise obligatory in the absence of *di*.

(84)

War	na=ñu={**ko**}	[di={**ko**}	jox	xaalis	].
must	C=3SG.S=3SG.O	DI=3SG.OBJ	give	money	

'We must give her/him money.' (Martinović, 2024, p.1596)

However, the optional presence of *di* seemingly has no discernible influence on meaning. Two examples are shown in (85) and (86).

(85)

Da=ma	jéem	[(di)	jox	xaalis	samay	wayjur.
do.C=1SG.S	try	DI	give	money	POSS.1PL	parents

'I tried giving money to my parents.'

(86)

War	na=ñu	[(di)	jox	miskin	yi	xaalis	bës	bu	nekk].
must	C=1PL.S	DI	give	poor	the.PL	money	day	C_{rel}	exist

'We must give money to the poor every day.'

While it is not our goal in this paper to provide a full analysis of restructuring verbs and their complements, we sketch here how the analysis of *di* that we developed in section 6 is compatible with data like (85)-(86). We propose that the complement clauses contain one of the covert prospective or imperfective operators to deliver a future or present temporal orientation of the matrix predicate. We have said that the prospective and imperfective must be licensed by *di*. In cases like (85) and (86), we suggest that these operators can be licensed by other modal elements, which we suggest are the embedding verbs in these cases. Thus, *di* is not required in these clauses as a licensor for these covert operators, and its optional presence simply adds an additional vacuous layer of modality. This data also provides evidence that *di* does not contribute the prospective and habitual meanings on its own, lending support to our analysis that these meanings are factored out of *di* to silent operators.

By contrast, *di* is obligatory with the restructuring predicate *wey* 'continue', as in (87).

(87)

Astu	wéy	na=∅	[*(di)	lekk	suukar].
Astou	continue	C=3SG.S	DI	eat	sugar

'Astou continued eating sugar (e.g., even after her doctor forbade it).'

We suggest here that the aspectual verb *wéy* 'continue' does not contribute any modality, and therefore cannot license the covert aspectual operators on its own. In this case, *di* is required to license the covert prospective aspect to derive the future orientation of the subordinate clause.

8 Conclusions

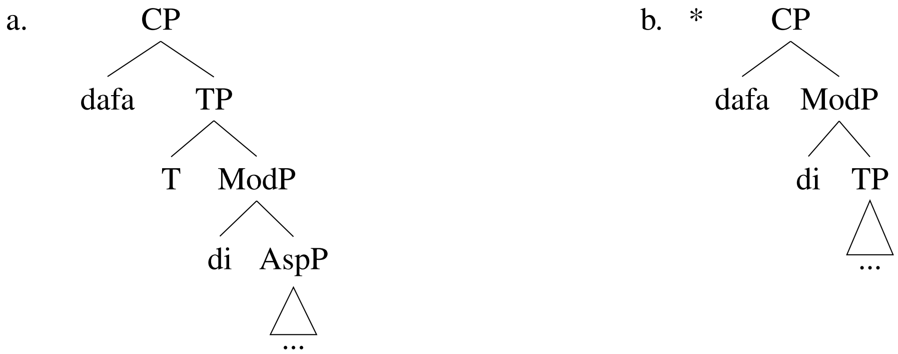
We began with a puzzle involving the many uses of the Wolof auxiliary *di* and restrictions on those uses depending on *di*'s syntactic position, with a view towards a unified analysis. In our account, we took advantage of existing analyses of event-based modality designed to capture (i) restrictions on modal flavour based on syntactic position (Hacquard, 2006, 2010), and (ii) modal properties of progressives and habituals (Portner, 1998; Ferreira, 2016). In doing so, we were also able to capture a curious feature of the future uses of *di*, namely that the use of high vs. low *di* results in different flavours of predictions. These distinct flavours of the future fall out from our analysis correlating syntactic position and type of modal base, and provide new insights into the modal flavour(s) of the future, which has been the subject of some debate in the literature.

Given that we propose that the two different readings of *di* result from two different positions of the Mod head in the clausal spine, we do not account for the fact that clauses in which a verbal element moves to C can only contain Mod above the TP, and clauses in which the verb stays below C can only have Mod below TP. In our analysis, the complement of *na* can only be as in (88a), and not as in (88b). Since in our analysis the high and low position of *di* is not derived through head movement, we currently cannot explain why *na* cannot take as a complement a TP that contains the low *di*.

(88)

Conversely, we also cannot account for the fact that the complementizer *dafa* only occurs with a low *di*, as in (89a), and not with high *di*, as in (89b). (89b) could be excluded through selectional restrictions (e.g., *dafa* cannot select for ModP)—though this is a descriptive, not an explanatory analysis—but we have no way to exclude (88b).

(89)



Another property of the clauses investigated in this paper that we have not discussed is their information structure. Descriptively, clauses where C is spelled out as *na* are used in two contexts: when the whole sentence contains new information, and when the whole sentence contains old information/information that is given in the context (e.g., I was expecting something to happen and now it has, so I am informing you of it using a *na*-sentence). Clauses where C is spelled out as *dafa* are used in predicate focus and so-called *explicative* contexts – they are good answers to *why*-questions. Additionally, speakers often report that the use of a *dafa*-sentence involves a greater degree of certainty, an effect similar to that we found in the future uses of *di*. It is possible that the restriction to 'high *di*' in *na*-sentences and to 'low *di*' in *dafa*-sentences is related to their information-structural properties. Seeing how no formal analysis of information structure in Wolof exists, we leave this issue for future work.

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